

IILM UNIVERSITY, GREATER NOIDA		LABORATORY MANUAL
PRACTICAL EXPERIMENT INSTRUCTION SHEET		
EXPERIMENT TITLE: Determination of energy Band Gap of a Semiconductors using Four-Probe method.		
EXPERIMENT NO. 01	DEPT. : Physics	ISSUE DATE: 12 August, 2026
LABORATORY: 26SOSP101 Semiconductor and Quantum Physics Lab	SEMESTER: I, II	TOTAL PAGE: 06

AIM: Determination of energy Band Gap of a Semiconductors using Four-Probe method.

APPARATUS: Measurement unit, Oven arrangement, Four probe arrangement.

THEORY AND FORMULA USED:

Energy band gap of semiconductor:

$$E_g = 2k \frac{2.3026 \times 2k \log_{10} \rho}{1/T} \text{ in eV}$$

where k is Boltzmann constant equal to $8.6 \times 10^{-5} \text{ eV/K}$, and ρ is the resistivity of the semiconductor crystal; given by

$$\rho = \frac{\rho_0}{f(w/s)};$$

where, $\rho_0 = 2\pi s (V/I)$,

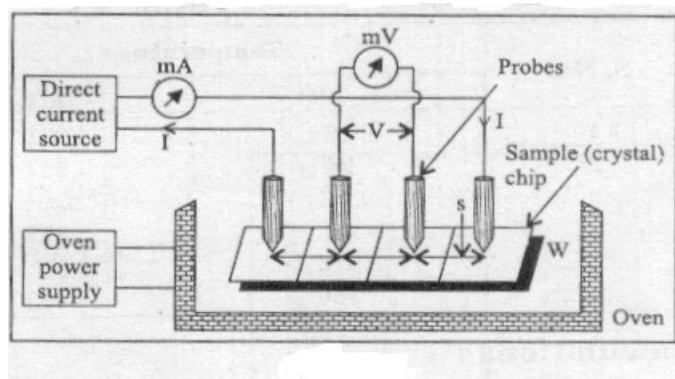
f(w/s): correction factor

s : distance between probes

w: thickness of semiconducting crystal

V and I are the voltage and current across and through the crystal chip.

CIRCUIT DIAGRAM:



PROCEDURE:

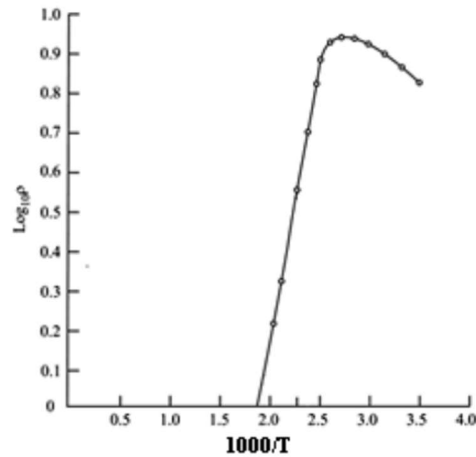
1. Connect the circuit as shown in figure.
2. In the four probes connect the outer probe leads to the current source in supply and inner probe leads to millivoltmeter.
3. Switch on the constant current source (keeping the oven OFF) and adjust current I to a desired value, 5 mA.
4. Place the four probe arrangement in the oven.
5. Place the thermometer close to the semiconductor crystal to measure the variation of temperature of crystal from room temperature to 120°C (in steps of 5°C).
6. Connect the oven power supply and start heating.
7. Measure the inner probe voltage V for various temperatures during heating or cooling.

OBSERVATIONS:

1. Distance between the probes, ' s ' = 0.2 cm
2. Constant current, ' I ' = 5 mA
3. Thickness of the semiconductor crystal, ' w ' = 0.05 cm
4. $f(w/s) = 5.89$

OBSERVATION TABLE:

S.No	Temperature			Voltage (mV)	ρ (ohm.cm)		log ρ
	in $^{\circ}\text{C}$	in K	1000/T		Observed ρ_0 $\rho_0 = 2\pi VS/l$	Corrected $\rho = \frac{\rho_0}{f\left(\frac{w}{s}\right)}$	
1							
2							
3							
4							
.....							

GRAPH:

Above is an ideal graph for band gap but practically it varies due to some variables such as crystal structure, temperature etc.

CALCULATION:

1. Find resistivity ρ_0 of semiconductor for different values of V & I, which give resistivity at various temperatures (This is the resistivity of the crystal of infinite dimensions. As the dimensions of the crystal used are finite we have to apply the correction to find resistivity by dividing the ρ_0 by $f(w/s)$, which is given in table.

$$\rho_0 = 2\pi sV/I = \text{-----ohm.cm}$$

2. Find w/s and then corresponding value of $f(w/s)$ from the table.
3. After choosing $f(w/s)$, calculate the value for resistivity For the various ρ_0 by using the formula

$$\rho = \frac{\rho_0}{f\left(\frac{w}{s}\right)} .$$

4. Plot the graph between $\log \rho$ along Y- axis and $1000/T$ along X- axis.
5. Find the slope from the graph for calculation

$$\frac{AB}{BC} = \frac{\log \rho}{1000/T}$$

So the energy band gap of semiconductor (Ge or Si) is

$$E_g = 2k \frac{2.3026 \times 2k \log_{10} \rho}{1/T} \times \frac{1000}{1000} \text{ in eV}$$

$$= 2 \times 8.6 \times 10^{-5} \times 2.3026 \times 1000 \times \frac{AB}{BC} \text{ eV}$$

$$E_g = 0.396 \times \text{Slope eV}$$

RESULT:

Energy band gap for semiconductor $E_g = \text{----- eV}$

PERCENTAGE ERROR:

$E_g (\text{Ge}) = 0.7 \text{ eV}$, and

$E_g (\text{Si}) = 1.12 \text{ eV}$

$$\text{Percentage error} = \frac{E_{\text{standard}} - E_{\text{calculated}}}{E_{\text{standard}}} \times 100 = \text{.....}$$

PRECAUTIONS:

1. The resistivity of the material should be uniform in the area of measurement.
2. Surface of the semiconductor chip should be flat.
3. Tip of the probes should very small in diameter compared to the distance between the probes.

VIVA QUESTIONS

1. Why is Four-Probe method preferred over other conventional methods for measuring resistivity?
2. Why is the current kept constant for measuring the resistivity of a semiconductor using four probe at different temperatures?
3. How do the conductivities of metals and semiconductor depend on temperature?
4. What is the equation giving the variation of conductivity/resistivity of a semiconductor with temperature?
5. For calculating the band gap from resistivity measurements why is it recommended to heat the sample over 100 degree Celsius?
6. What are the values of band gaps for Silicon and Germanium?
7. How does the band gap of a semiconductor vary with temperature?